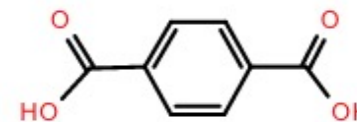


The Conventional Route: Solvothermal Synthesis

ZnCl₄

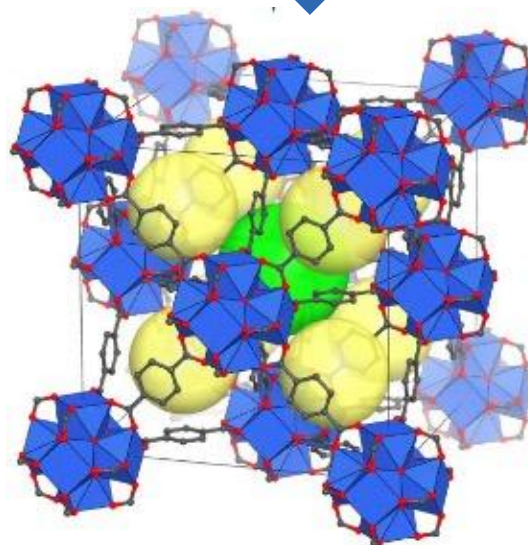
Simple metal salt



Organic linker

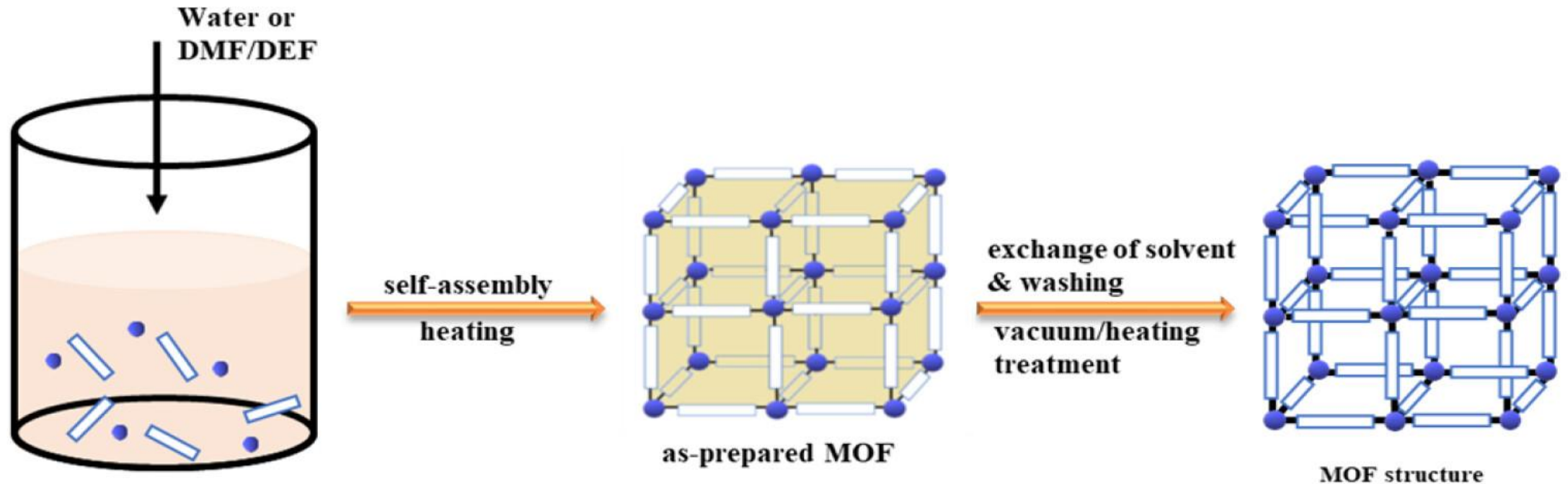
1. Sonicate in high-boiling solvents (e.g., DMF).
2. Heating for hours/ days
3. Centrifuge and filter

Optimization: Controlling temperature, order of addition, and metal source.



The Role of Modulators: Benzoic or acetic acids slow nucleation to promote ideal framework assembly.

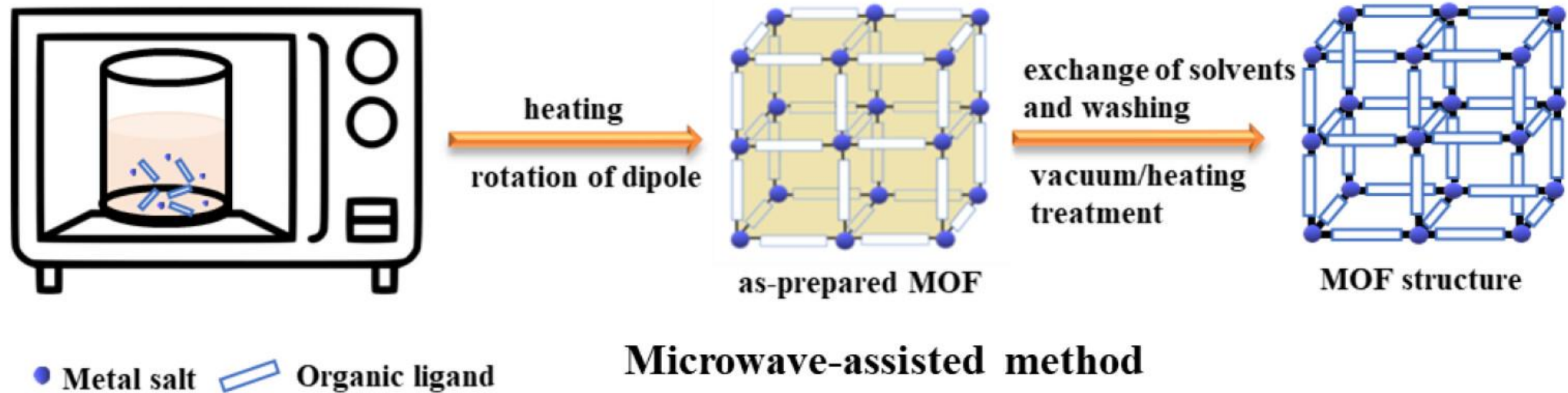
The Conventional Route: Solvothermal Synthesis



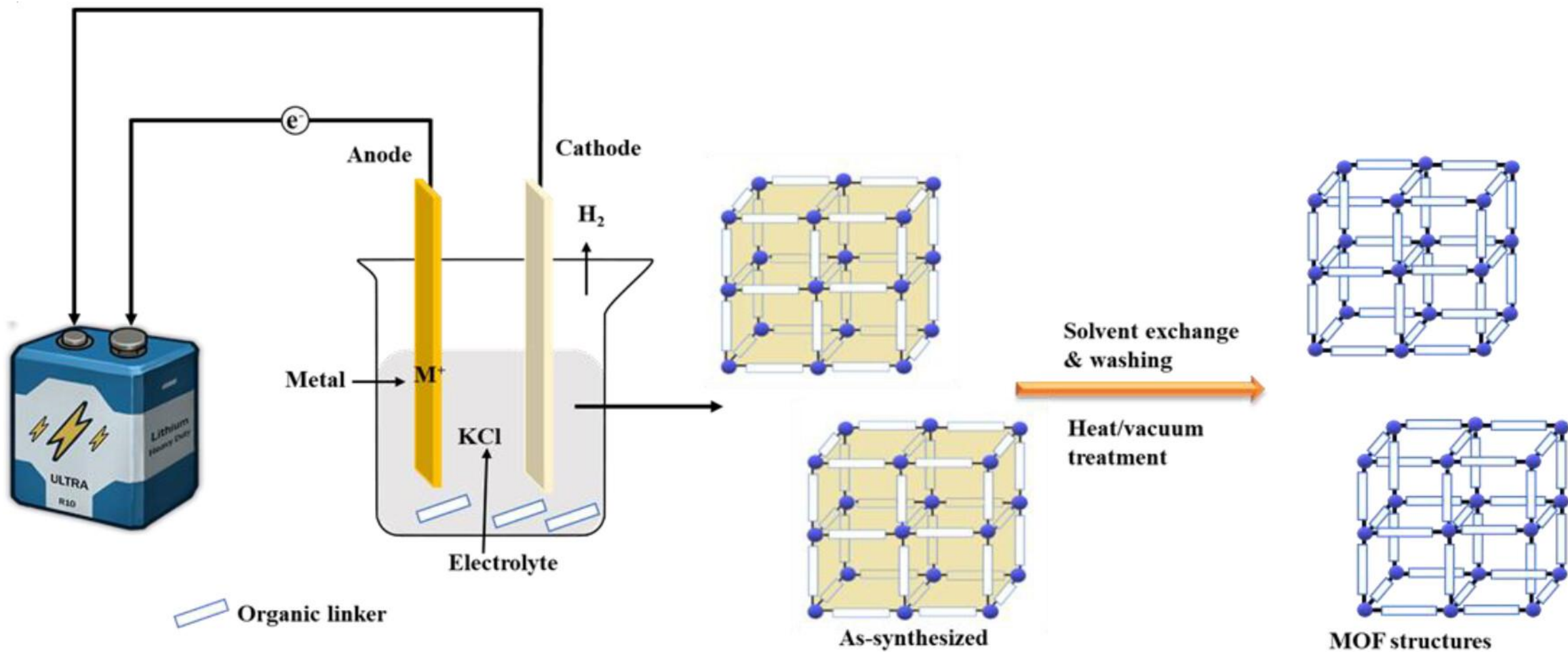
● Metal salt ▭ Organic ligand

Solvothermal method

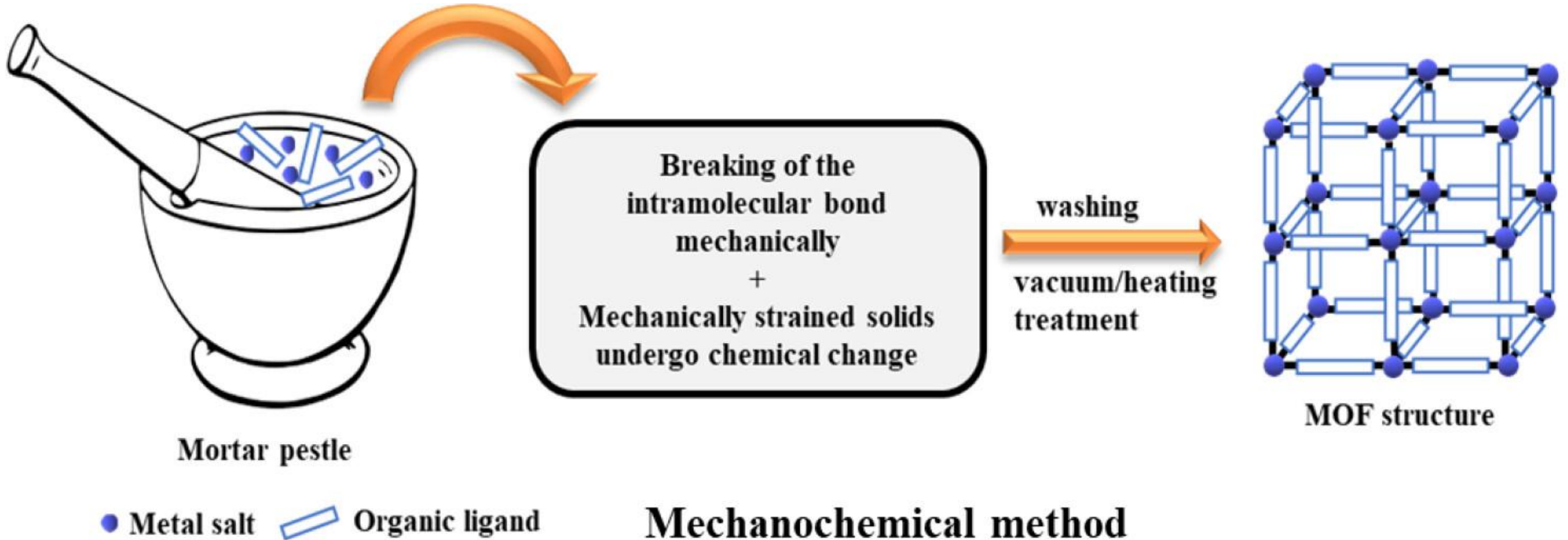
The Conventional Route: Microwave-Assisted Method



Electrochemical Method



Mechanochemical Method



Sonochemical Method

